

Atividade 3 - Sistemas Dinâmicos Contínuos

Questão 01:

Condições iniciais: $y(0) = 1$; $\dot{y}(0) = 0$

$$\ddot{y}(t) + 5\dot{y}(t) + 6y(t) = x(t)$$

$n=2$

$$D^2 y(t) + 5Dy(t) + 6y(t) = x(t)$$

$m=1$

$$D^2 y_0(t) + 5Dy_0(t) + 6y_0(t) = 0 \quad \text{Entrada nula}$$

$$(D^2 + 5D + 6)y_0(t) = 0 \rightarrow Q(D)y_0(t) = 0$$

$$(\lambda^2 + 5\lambda + 6)y_0(t) = 0 \therefore (\lambda^2 + 5\lambda + 6) \text{ é o pol. nom. o característico}$$

$$\lambda^2 + 5\lambda + 6 \quad \lambda = \frac{-5 \pm \sqrt{25 - 24}}{2} = \frac{-5 \pm 1}{2} \therefore \lambda_1 = \frac{-5 + 1}{2} = -2$$

$$\Delta = b^2 - 4ac = 25 - 24 = 1$$

$$\lambda_2 = \frac{-6}{2} = -3$$

$$y_0(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} = 0$$

RAÍZES CARACTERÍSTICAS

$$y(0) = c_1 e^{-2t} + c_2 e^{-3t} = 1$$

$$\frac{dy_0(t)}{dt} = c_1 \lambda_1 e^{\lambda_1 t} + c_2 \lambda_2 e^{\lambda_2 t} = 0$$

$$\dot{y}(0) = -2c_1 e^{-2t} - 3c_2 e^{-3t} = 0$$

$$\begin{cases} c_1 e^{-2t} + c_2 e^{-3t} = 1 & (1) \rightarrow 2c_1 e^{-2t} + 3c_2 e^{-3t} = 2 \\ -2c_1 e^{-2t} - 3c_2 e^{-3t} = 0 \end{cases}$$

$$-c_2 e^{-3t} = 2 \rightarrow -c_2 e^{-3t} = 2 \therefore c_2 = -2$$

$$c_1 \cdot 1 + (-2) \cdot 1 = 1 \rightarrow c_1 - 2 = 1 \quad c_1 = 3$$

$$3e^{-2t} - 2e^{-3t} = y(t) \quad \text{Resposta a entrada nula}$$

Questão 02:

condições iniciais $y(0) = 1$; $\dot{y}(0) = 0$

$$\ddot{y}(t) + 2\dot{y}(t) + y(t) = x(t)$$

$$(D^2 + 2D + 1)y_0(t) = 0$$

$$(\lambda^2 + 2\lambda + 1)y_0(t) = 0 \therefore (\lambda^2 + 2\lambda + 1) \text{ é o polinômio característico}$$

$$\lambda^2 + 2\lambda + 1 = 0 \quad \lambda = \frac{-2 \pm \sqrt{0}}{2} = \frac{-2}{2} = -1 \quad \lambda_1 = \lambda_2$$

$$\Delta = 4 - 4 = 0$$

$$y_0(t) = (c_1 + c_2 t) e^{\lambda t} = (c_1 + c_2 t) e^{-t}$$

A multiplicidade

é igual a 2 (pois),

$$y_0(0) = (c_1 + c_2 \cdot 0) e^0 = c_1 = 1$$

pois é a quantidade que a matriz coincide em

$$\frac{d}{dt}(u \cdot v) = u \frac{dv}{dt} + v \frac{du}{dt} \rightarrow u = (c_1 + c_2 t)$$

$$\frac{du}{dt} = 0 + c_2 = c_2$$

$$Dy_0(t) = (c_1 + c_2 t) \cdot e^{-t} + e^{-t} \cdot c_2$$

$$Dy_0(t) = (c_2) e^{-t} + (c_1 + c_2 t) (-e^{-t}) \quad v = e^{-t}$$

$$Dy_0(t) = e^{-t} [c_2 + (c_1 + c_2 t) (-1)] \quad \frac{dv}{dt} = -e^{-t}$$

$$Dy_0(t) = e^{-t} [c_2 - c_1 - c_2 t]$$

$$\dot{y}_0(0) = e^0 [c_2 - c_1 - c_2 \cdot 0] = 1 [c_2 - c_1] = 0$$

$$\therefore c_2 - c_1 = 0 \rightarrow c_2 = c_1$$

$$(1 + 1t) e^{-t} = y_0(t)$$

Resposta Nome: Gerson

Questão 03:

CONDIÇÕES INICIAIS $y(0) = 1$; $\dot{y}(0) = 0$

$$\ddot{y}(t) + 9y(t) = x(t)$$

$$(D^2 + 9)y_0(t) = 0$$

$$(\lambda^2 + 9)y_0(t) = 0 \quad (\lambda^2 + 9) \text{ é o polinômio característico}$$

$$\lambda^2 + 9 = 0 \rightarrow \lambda^2 = -9 \rightarrow \lambda = \pm \sqrt{-9} \rightarrow \lambda = \pm \sqrt{9} \cdot \sqrt{-1}$$

$$\lambda = \pm 3j \quad \text{Raízes características}$$

$$y_0(t) = C e^{at} \cos(bt + \theta) = C e^0 \cos(3t + \theta) = C \cos(3t + \theta)$$

$$\dot{y}_0(t) = -C \sin(3t + \theta) \cdot 3 = -3C \sin(3t + \theta)$$

$$y_0(0) = C \cos(3 \cdot 0 + \theta) = C \cos(\theta) = 1$$

$$\dot{y}_0(0) = -3C \sin(\theta) = 0$$

$$C = 0 \rightarrow \text{solução trivial} \quad (\text{NÃO USAR}) \quad \therefore \sin(\theta) = 0$$

$$\theta = k\pi = 0$$

$$C \cos(\theta) = 1 \rightarrow C = 1$$

$$y_0(t) = \cos(3t) \quad \text{Solução homogênea na forma senoidal}$$

$$\cos(p) = \frac{1}{2} (e^{ip} + e^{-ip}) \quad \text{fórmula de Euler}$$

$$\cos(3t) = \frac{1}{2} (e^{3jt} + e^{-3jt})$$

$$y_0(t) = \frac{1}{2} e^{3jt} + \frac{1}{2} e^{-3jt}$$

$$c_1 = 1/2 \quad c_2 = 1/2$$

Questão 01:

CONDICÕES INICIAIS $y(0) = 1$; $\dot{y}(0) = 0$

$$\ddot{y}(t) + 2\dot{y}(t) + 5y(t) = x(t)$$

$$(D^2 + 2D + 5)y_0(t) = 0$$

$$(\lambda^2 + 2\lambda + 5)y_0(t) = 0$$

$$\lambda^2 + 2\lambda + 5 \quad \lambda = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4j}{2} = (-1 \pm 2j)$$

$$\Delta = 4 - 20 = -16$$

$$\frac{-2 - 4j}{2} = (-1 - 2j)$$

$$\lambda_1 = -1 + 2j$$

$$\lambda_2 = -1 - 2j \quad \text{RAÍZES DO POLINÔMIO CARACTERÍSTICO (CONJUGADO COMPLEXO)}$$

$$y_0(t) = C e^{-t} \cos(2t + \theta)$$

$$u = C e^{-t}$$

$$\frac{du}{dt} = -C e^{-t}$$

$$\dot{y}_0(t) = (C e^{-t}) (-2 \sin(2t + \theta)) +$$

$$(\cos(2t + \theta)) (-C e^{-t})$$

$$\theta = \cos(2t + \theta)$$

$$\dot{y}_0(t) = C e^{-t} \cdot 2 \sin(2t + \theta) - C e^{-t} \cdot \cos(2t + \theta) \frac{d\theta}{dt} = -2 \sin(2t + \theta)$$

$$(-C e^{-t}) [2 \sin(2t + \theta) + \cos(2t + \theta)]$$

$$\rightarrow C \cdot \frac{2}{5} = 1 \Rightarrow C = \frac{5}{2}$$

$$y_0(0) = C e^0 \cos(2 \cdot 0 + \theta) = C \cos(\theta) = 1$$

$$\dot{y}(0) = -C e^0 [2 \sin(\theta) + \cos(\theta)] = -C [\cos(\theta) + 2 \sin(\theta)]$$

$$[\cos(\theta) + 2 \sin(\theta)] = 0 \quad (\cos(\theta)) \neq 1 + 2 \tan(\theta) = 0$$

$$\tan(\theta) = \frac{-1}{2}$$

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

$$\arctan\left(\frac{-1}{2}\right) \approx -0,46 \text{ rad} = \theta \quad \cos(\theta) = \frac{2}{5}$$

$$y^2 + 2^2 + 5$$

exponencial / Real

$$y_0(t) = \frac{5}{2} e^{-t} \cos\left(2t + \arctan\left(\frac{-1}{2}\right)\right)$$

Questão 05: Condições Iniciais: $y(0) = 1$, $\dot{y}(0) = 0$, $\ddot{y}(0) = 0$

$$\ddot{y}(t) + 6\dot{y}(t) + 10y(t) + 8y(t) = x(t)$$

$$(D^3 + 6D^2 + 10D + 8)y_0(t) = 0$$

$$\lambda^3 + 6\lambda^2 + 10\lambda + 8 = 0$$

$$\lambda(\lambda^2 + 6\lambda + 10) + 8 = 0 \quad \lambda(\lambda^2 + 6\lambda + 10) = -8 \quad \lambda_2 = 0$$

Inspeção (-4)

$$-4(16 - 24 + 10) = -4(2) = -8 \quad \therefore (\div (\lambda + 4))$$

$$\lambda^3 + 6\lambda^2 + 10\lambda + 8 = (\lambda + 4)(\lambda^2 + 2\lambda + 2) = 0 \quad \lambda_1 = -4$$

$$\lambda^2 + 2\lambda + 2 = 0 \quad \lambda = \frac{-2 \pm \sqrt{4 - 8}}{2} = \frac{-2 \pm 2i}{2} \quad \lambda_1 = -1 + i$$

$$\Delta = 4 - 4 \cdot 2 = -4 \quad \lambda_2 = -1 - i$$

Elaborar a partir das raízes conclua V

$$y_0(t) = C_1 e^{-4t} + e^{-t} [C_2 \cos(\beta t) + C_3 \sin(\beta t)] \quad \Delta e^{\sigma t} \cos(\beta t + \theta)$$

$$y_0(t) = C_1 e^{-4t} + e^{-t} [C_2 \cos(t) + C_3 \sin(t)] \quad (\text{normalizada})$$

$$y_0(t) = C_1 e^{-4t} + C_2 e^{-t} \cos(t) + C_3 e^{-t} \sin(t)$$

$$\dot{y}_0(t) = -4C_1 e^{-4t} + e^{-t} [(-C_2 + C_3) \cos(t) + (-C_2 - C_3) \sin(t)]$$

$$\ddot{y}_0(t) = 16C_1 e^{-4t} + e^{-t} [(-2C_3) \cos(t) + (2C_2) \sin(t)]$$

$$y_0(0) = 1 \rightarrow y_0(t) = C_1 + C_2 + C_3 \cdot 0 = C_1 + C_2 = 1$$

$$\dot{y}_0(0) = 4C_1 + C_2 - C_3 = 0$$

$$\ddot{y}_0(0) = 8C_1 = C_3$$

$$4C_1 + C_2 - (8C_1) = 0 \rightarrow C_2 - 4C_1 = 0 \rightarrow C_2 = 4C_1$$

$$C_1 + (4C_1) = 1 \quad \therefore C_1 = 1/5$$

$$C_2 = 4C_1 = \frac{4}{5} \quad C_3 = 8C_1 = \frac{8}{5}$$

$$y_0(t) = \frac{1}{5} e^{-4t} + e^{-t} \left[\frac{4}{5} \cos(t) + \frac{8}{5} \sin(t) \right]$$